CO2 - AT3

80%

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ITA0517 - Computer Vision

Class Test

20/25

1) Histogram processing for excessive brightness variation

2) Grey level transformation for subtle intensity differences

3) Salt - & - pepper noise removal

4) Restoring details after excessive smoothing

5) filtering while preserving important structures

1 sol^n :- Histogram Equalization

Histogram equalization is used to improve the contrast of an image when the intensity values are unevenly distributed

Working

1. Calculate the histogram of the image
2. find the cumulative distribute function
3. Use the CDF to map the original intensity values to new values
4. This spreads the intensity values over a wider range

Justification

when different regions have different brightness levels some details may be clearly visible histogram equalization redistributes the intensity levels & ~~improves~~ improves overall contrast

Expected outcome

The image has a more balanced intensity distribution with better contrast & improved visibility of details especially in dark or bright regions.

2) Sol^n:- Contrast stretching

Contrast stretching is a grey level transformation used to enhance small differences in intensity.

Working

- The original intensity range is identified
- Low intensity values are mapped to lower value & high intensity values to high values
- Intermediate values are stretched over a wider range

$$s = \frac{(r - r_{min})}{(r_{max} - r_{min})} \times 255$$

r is the original intensity

s is the new intensity

Justification

Medical images often contain subtle differences b/w tissues. Contrast stretching increases these differences without changing the basic structure

Effect on diagnosis

Fine structure & small intensity variations become more visible, helping doctors identify abnormalities more easily.

3) Median filter

A median filter is the most appropriate spatial filter for salt & pepper noise

Working

1. A small window such as 3×3 is moved across the image
2. Pixel values inside the window are arranged in ascending order
3. The middle value (median) replaces the center pixel.
4. The process is repeated for all pixels

Justification

Unlike averaging filter, the median filter moves isolated black and white noise while preserving edge

~~4) Unsharp filter masking~~

5) Bilateral filtering

Bilateral filtering is ~~very~~ useful when image contains noise but important edge & structure must be preserved.

Working

- The original intensity range is identified
- Pixel values inside the window are arranged in ascending order.

- Reduces high-frequency noise
- Reduces Blur edges

Justification

- A normal Gaussian/mean reduce noise but also blur important edge. Bilateral filtering reduces high-frequency noise while preserving significant

4) Unsharp Masking

- Image enhancement for edges we can use technique called "unsharp Masking" (or) "High-boost filtering"
- Working
- It blurs the original image by using Gaussian method
- Subtracts the blurred image from the original image to obtain the edges clearly.
- The obtained edges will be added to original image

Process:

Original image
↓
Gaussian filter
↓
Blured imge
↓
Original − Blured image
↓
Edge Detection
↓
Add unsharp Masking Algorithm
↓
Sharpened Image

Justification

So, by using unsharp Masking algorithm we can find edge detection. when it will help to image enhancement for edges